

questions

information on Computer Network Security

Sec on 2 – Prac ce Ques ons

Name: fares abdelrahman

ID: 2302590

Sec on: s7

Part 1: Modular Arithmetic

1- $15 \bmod 7 = 15 = 7 \times 2 + 1 \implies 1$ الباقي

2- $23 \bmod 5 = 23 = 5 \times 4 + 3 \implies 3$ الباقي

3- $42 \bmod 8 = 42 = 8 \times 5 + 2 \implies 2$ الباقي

Q2

1. 9 and 16 $\Rightarrow 16 = 7 \times 2 + 2$, الباقي 2
 $9 = 7 \times 1 + 2$, الباقي 2

اذن ال 16 و 9 متكافين

2. -5 and 2 $\Rightarrow -5 = -5 + 7 = 2$, الباقي 2
 $2 = 2 \bmod 7 = 2$ الباقي 2

اذن ال -5 و 2 متكافين

3. 14 and 3 $\Rightarrow 14 = 7 \times 2 + 0$, الباقي 0
 $3 = 3 \bmod 7 = 3$, الباقي 3

اذن ال 14 و 3 مش متكافين

Q3

1. $35 \bmod 6 \Rightarrow 35 = 6 \times 5 + 5 \Rightarrow 5$

2. $50 \bmod 9 \Rightarrow 50 = 9 \times 5 + 5 \Rightarrow 5$

3. $-8 \bmod 5 \Rightarrow -8 + 5 \Rightarrow -3 + 5 = 2 \Rightarrow 2$

Q4

1. $3 \bmod 7 \Rightarrow 3 \times 1 \bmod 7 = 3$

$\Rightarrow 3 \times 2 \bmod 7 = 6$

$\Rightarrow 3 \times 3 \bmod 7 = 2$

$\Rightarrow 3 \times 4 \bmod 7 = 5$

$\Rightarrow 3 \times 5 \bmod 7 = 1 \Rightarrow \text{the inverse of 3 is 5}$

2. $4 \bmod 7 \Rightarrow 4 \times 1 \bmod 7 = 4$

$\Rightarrow 4 \times 2 \bmod 7 = 1 \Rightarrow \text{the inverse of 4 is 2}$

Q5

1. $4 \bmod 8$ not exists because $\gcd(4,8) \neq 1$

2. $5 \bmod 11$ exists because $\gcd(5,11) = 1$

the inverse $\Rightarrow 5 \times 1 \bmod 11 = 5$

$$\Rightarrow 5 \times 2 \bmod 11 = 10$$

$$\Rightarrow 5 \times 3 \bmod 11 = 4$$

$$\Rightarrow 5 \times 4 \bmod 11 = 9$$

$$\Rightarrow 5 \times 5 \bmod 11 = 3$$

$$\Rightarrow 5 \times 6 \bmod 11 = 8$$

$$\Rightarrow 5 \times 7 \bmod 11 = 2$$

$$\Rightarrow 5 \times 8 \bmod 11 = 7$$

$$\Rightarrow 5 \times 9 \bmod 11 = 1$$

Q6

Find the multiplicative inverse of:

$$7 \bmod 10$$

$$\Rightarrow 7 \times 1 \bmod 10 = 7$$

$$\Rightarrow 7 \times 2 \bmod 10 = 4$$

$$\Rightarrow 7 \times 3 \bmod 10 = 1 \Rightarrow \Rightarrow \Rightarrow \text{the inverse is } 3$$

Q7

Find the GCD of the following numbers:

1. $\gcd(30,12) \Rightarrow 30=2 \times 3 \times 5 \quad ,, \quad 12=2 \times 2 \times 3 \quad \Rightarrow 2 \times 3 = 6 \text{ gcd } 6$

2. $\gcd(84,30) \Rightarrow 84= 2 \times 2 \times 3 \times 7 \quad ,, \quad 30=2 \times 3 \times 5 \quad \Rightarrow 2 \times 3 = 6 \text{ gcd } 6$

3. $\gcd(54,24) \Rightarrow 54=2 \times 3 \times 3 \times 3 \quad ,, \quad 24=2 \times 2 \times 2 \times 3 \Rightarrow 2 \times 3 = 6 \text{ gcd } 6$

Q8

Find the GCD using prime factorization:

$$1. \gcd(60, 48) \Rightarrow 60 = 2 \times 2 \times 3 \times 5$$

$$\Rightarrow 48 = 2 \times 2 \times 2 \times 2 \times 3$$

$$\Rightarrow \gcd = 2 \times 2 \times 3 \Rightarrow 12$$

$$2. \gcd(72, 90) \Rightarrow 72 = 2 \times 2 \times 2 \times 3 \times 3$$

$$\Rightarrow 90 = 2 \times 3 \times 3 \times 5$$

$$\Rightarrow \gcd = 2 \times 3 \times 3 \Rightarrow 18$$

Q9

Use the Euclidean Algorithm to find:

1. $\gcd(27, 21) \Rightarrow \gcd(21 \times 1 + 6, 21) \Rightarrow \gcd(21, 6)$
 $\Rightarrow \gcd(6 \times 3 + 3, 6) \Rightarrow \gcd(6, 3)$
 $\Rightarrow \gcd(3 \times 2 + 0, 3) \Rightarrow \gcd(3, 0) \Rightarrow \gcd = 3$
2. $\gcd(270, 192) \Rightarrow \gcd(192 \times 1 + 78, 192) \Rightarrow \gcd(192, 78)$
 $\Rightarrow \gcd(78 \times 2 + 36, 78) \Rightarrow \gcd(78, 36)$
 $\Rightarrow \gcd(36 \times 2 + 6, 36) \Rightarrow \gcd(36, 6)$
 $\Rightarrow \gcd(6 \times 6 + 0, 6) \Rightarrow \gcd(6, 0) \Rightarrow \gcd = 6$

Q10

Use the Euclidean Algorithm to compute:

$$\begin{aligned}\gcd(741, 285) &\Rightarrow \gcd(285 \times 2 + 171, 285) \Rightarrow \gcd(285, 171) \\ &\Rightarrow \gcd(171 \times 1 + 114, 171) \Rightarrow \gcd(171, 114) \\ &\Rightarrow \gcd(114 \times 1 + 57, 114) \Rightarrow \gcd(114, 57) \\ &\Rightarrow \gcd(57 \times 2 + 0, 57) \Rightarrow \gcd(57, 0) \Rightarrow \gcd = 57\end{aligned}$$

Q11

Explain why a multiplicative inverse does not exist for:

$4 \bmod 8$

شرط وجود المعكوس الضربي لـ a و b لازم يكون الـ

$$\gcd(a,b)=1$$

$$\gcd(4,8)=4$$

اذن الشرط لم يتحقق